

VSB ENGINEERING COLLEGE

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**DEPARTMENT OF ELECTRICAL AND ELECTRONICS
ENGINEERING**

PROJECT REPORT

ON

**"INTELLIGENT LIGHTING MANAGEMENT SYSTEM USING
ESP8266"**

Submitted by

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ABSTRACT

Energy conservation has become an important aspect of modern electrical and electronic systems due to the increasing demand for electricity and the need for efficient energy utilization. Conventional lighting systems are generally operated manually, which may result in unnecessary power consumption when lights remain switched ON during daylight or in well-lit environments. To overcome this problem, an automatic lighting control system is an effective and economical solution.

This project, "**Intelligent Lighting Management System Using ESP8266**," is designed to automatically control lighting based on the surrounding light intensity. The system uses an **ESP8266 NodeMCU** microcontroller and an **LDR (Light Dependent Resistor) sensor module** to detect ambient light conditions. The LDR continuously senses the intensity of light and sends the corresponding signal to the ESP8266. Based on the sensor input, the microcontroller automatically switches the LED ON when the surrounding environment becomes dark and switches it OFF when sufficient light is available. The program is developed using the **Arduino IDE** and uploaded to the ESP8266 through a USB connection.

The hardware consists of an **ESP8266 NodeMCU**, **LDR sensor module**, **LED**, **220 Ω resistor**, **breadboard**, **jumper wires**, and a **USB power supply**. The system was assembled and tested under different lighting conditions to evaluate its performance. The results showed that the LED responded quickly and accurately to changes in ambient light intensity, ensuring reliable automatic operation. The project provides a simple, low-cost, and energy-efficient solution for intelligent lighting management.

This system can be further enhanced by integrating **IoT technology**, enabling remote monitoring and control through mobile applications or cloud platforms. It can be effectively applied in **homes, offices, educational institutions, industries, and street lighting systems**, contributing to energy conservation, reduced electricity costs, and the development of smart and sustainable environments.

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CHAPTER 1 – INTRODUCTION

1.1 Background

Energy conservation has become one of the most important concerns in today's world due to the increasing demand for electricity and the need to reduce energy wastage. Lighting systems consume a significant portion of electrical energy in homes, offices, educational institutions, industries, and public places. Conventional lighting systems are usually operated manually, which often leads to unnecessary power consumption when lights remain switched ON even during daylight or when adequate natural light is available.

With the rapid development of embedded systems and automation technologies, intelligent lighting systems have emerged as an effective solution for reducing energy consumption. The ESP8266 NodeMCU is a low-cost microcontroller with built-in Wi-Fi capability, making it suitable for developing smart automation projects. By interfacing an LDR (Light Dependent Resistor) sensor with the ESP8266, the system can continuously monitor ambient light intensity and automatically control lighting based on environmental conditions. This automation improves energy efficiency, minimizes human intervention, and provides a reliable and cost-effective lighting management solution.

1.2 Objectives

1. To develop an **Intelligent Lighting Management System** using the **ESP8266 NodeMCU**.
2. To interface an **LDR sensor module** with the ESP8266 for continuous monitoring of ambient light intensity.
3. To automatically switch the LED **ON** during dark conditions and **OFF** during bright conditions.
4. To reduce unnecessary electricity consumption through automatic lighting control.
5. To develop a simple, reliable, and low-cost embedded system for intelligent lighting applications.
6. To provide a system that can be enhanced with **IoT-based remote monitoring and smart home automation** in the future.

1.3 Scope of the Project

The proposed system is designed and implemented using an **ESP8266 NodeMCU**, an **LDR sensor module**, an **LED**, a **220 Ω resistor**, a **breadboard**, and **jumper wires**. The project focuses on detecting the surrounding light intensity and automatically controlling the lighting unit based on the sensor input. The system is programmed using the **Arduino IDE** and tested under different lighting conditions to ensure reliable operation.

The scope of this project includes sensor interfacing, automatic lighting control, hardware implementation, software development, and performance testing. The prototype serves as a low-cost and energy-efficient solution suitable for homes, offices, classrooms, corridors, and street lighting applications. The system can be further enhanced by integrating motion sensors, cloud-based monitoring, mobile application control, and solar-powered lighting systems, making it suitable for future smart city and IoT applications.

CHAPTER 2 – LITERATURE SURVEY

Conventional lighting systems are generally operated manually, requiring users to switch lights ON and OFF whenever needed. Although this method is simple, it often results in unnecessary electricity consumption because lights may remain ON even during daylight or when sufficient natural light is available. Such inefficient usage increases energy costs and contributes to excessive power consumption in residential, commercial, and public environments.

With the advancement of embedded systems and automation technologies, automatic lighting control systems have become an effective solution for improving energy efficiency. Light sensors such as **Light Dependent Resistors (LDRs)** are widely used to detect ambient light intensity because they are inexpensive, reliable, and easy to interface with microcontrollers. Many research studies and practical projects have combined LDR sensors with microcontrollers such as **Arduino**, **ESP8266**, and **ESP32** to develop intelligent lighting systems that automatically control lighting based on environmental conditions.

The **ESP8266 NodeMCU** has gained significant popularity in smart automation applications due to its low cost, compact size, built-in Wi-Fi capability, and ease of programming using the **Arduino IDE**. Compared to traditional microcontrollers, the ESP8266 offers better connectivity and can be integrated with IoT platforms for remote monitoring and control. Recent developments in smart lighting systems also incorporate cloud services, mobile applications, motion sensors, and energy monitoring features to provide more efficient and user-friendly solutions.

The present project focuses on designing and implementing a simple, reliable, and cost-effective **Intelligent Lighting Management System Using ESP8266** with an **LDR sensor module**. The system automatically controls the lighting unit according to the surrounding light intensity, reducing unnecessary electricity consumption while providing a foundation for future enhancements such as IoT-based monitoring, mobile application control, and smart city lighting systems.

CHAPTER 3 – SYSTEM DESIGN / METHODOLOGY

3.1 Block Diagram

The proposed system consists of a power supply stage, an LDR sensing stage, the ESP8266 NodeMCU processing stage, and an LED output stage. The LDR sensor continuously monitors the surrounding light intensity and sends the corresponding signal to the ESP8266. Based on the received input, the microcontroller automatically controls the LED by switching it ON during darkness and OFF during daylight.

Power Supply (5V USB) → LDR Sensor Module → ESP8266 NodeMCU (Decision Logic) → LED (Lighting Unit)

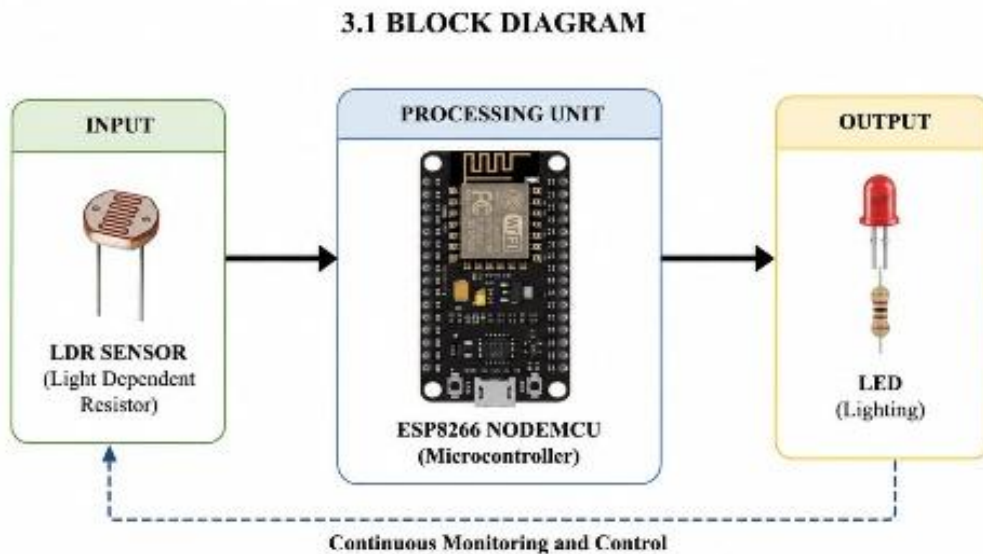


Fig. 3.1 Block Diagram of Intelligent Lighting Management System

Fig. 3.1: Block Diagram of the Intelligent Lighting Management System Using ESP8266

3.2 Flowchart

START → Initialize ESP8266 and GPIO Pins → Read LDR Sensor Value → Is Ambient Light Low? → YES: Turn LED ON → NO: Turn LED OFF → Wait for a Short Delay → Repeat

3.2 FLOWCHART

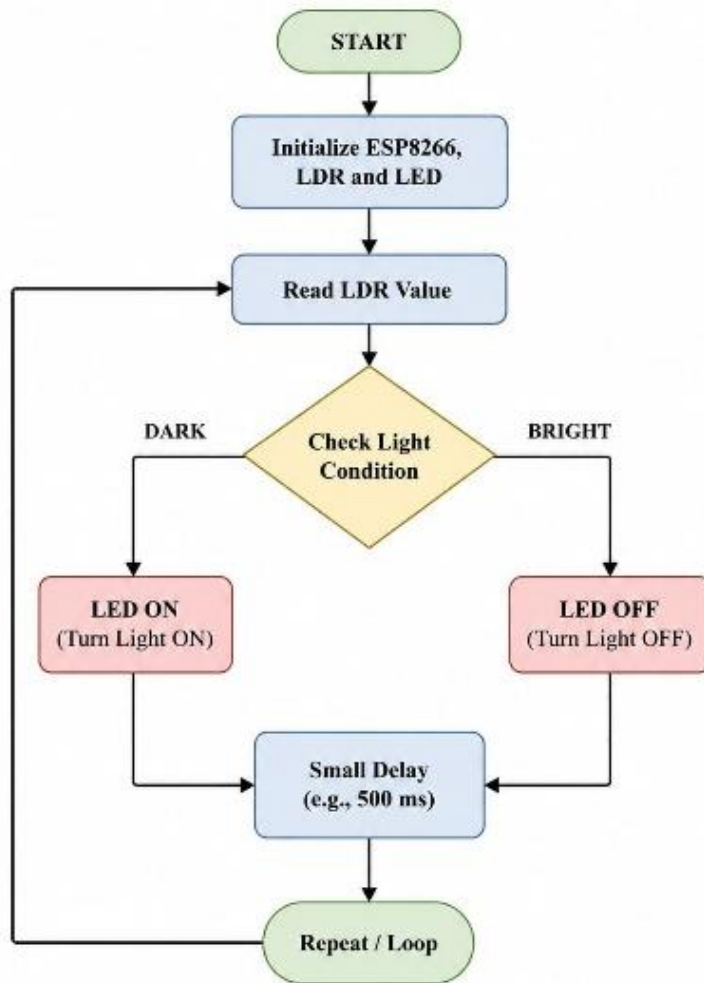


Fig. 3.2 Flowchart of Intelligent Lighting Management System using ESP8266 and LDR

Fig. 3.2: Flowchart of the Intelligent Lighting Management System

3.3 Hardware Requirements

S.No.	Component	Specification
1	ESP8266 NodeMCU	Wi-Fi Enabled Development Board
2	LDR Sensor Module	Digital Light Sensor Module
3	LED	5 mm White LED

4	Resistor	220 Ω Current Limiting Resistor
5	Breadboard	Solderless Breadboard
6	Jumper Wires	Male-to-Male Connecting Wires
7	USB Cable	Micro USB Cable for Programming and Power

Table 3.1: Hardware Components Used

3.4 Software Requirements

- Arduino IDE
- ESP8266 Board Package
- USB Driver for NodeMCU
- Serial Monitor for Program Testing and Debugging

3.5 Development Process

1. Studied the working principle of the **LDR sensor** and the **ESP8266 NodeMCU**.
2. Designed the block diagram and circuit connections for the ESP8266, LDR sensor, LED, and resistor.
3. Assembled the hardware components on a breadboard according to the circuit design.
4. Developed the program using the **Arduino IDE** for automatic lighting control.
5. Uploaded the program to the ESP8266 NodeMCU through a USB cable.
6. Tested the system under different lighting conditions to verify proper operation.
7. Observed the automatic switching of the LED based on ambient light intensity and confirmed the system's reliable performance.

CHAPTER 4 – IMPLEMENTATION

4.1 Project Development

The Intelligent Lighting Management System using ESP8266 was developed to automatically control lighting based on surrounding light intensity. The system uses an ESP8266 NodeMCU as the main controller and an LDR (Light Dependent Resistor) sensor module to detect ambient light levels.

The LDR sensor continuously monitors environmental light conditions and sends a digital signal to the ESP8266 NodeMCU. The microcontroller processes this input and controls the LED accordingly. When low light conditions are detected, the LED is switched ON automatically, and when sufficient light is available, the LED is turned OFF.

The hardware setup was assembled on a solderless breadboard using jumper wires. The program was developed using the Arduino IDE and uploaded to the ESP8266 NodeMCU via a Micro-USB cable. The system was tested under different lighting conditions and responded accurately, demonstrating efficient automatic lighting control and energy-saving capability.

4.2 Circuit Diagram / Pin Connections

ESP8266 NodeMCU Pin	Connected To	Function
3V3	LDR VCC	Power Supply
GND	LDR GND and LED Cathode	Common Ground
D1	LDR OUT	Light Sensor Input
D2	LED (through 220 Ω resistor)	Output Control

Table 4.1: Pin Connection Details

The circuit was implemented on a solderless breadboard. The ESP8266 NodeMCU was placed at the center for easy wiring. The LED was connected through a 220 Ω resistor to protect it from excess current. The entire system was powered using a Micro-USB cable.

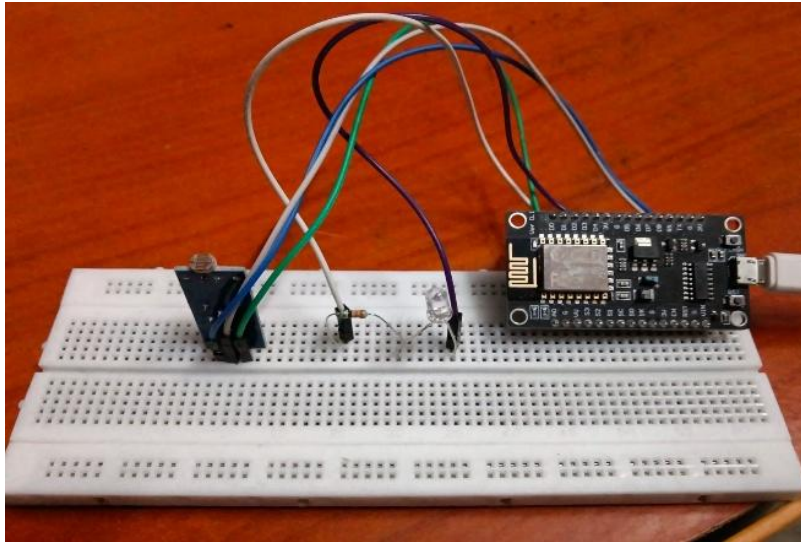


Figure 4.1: Circuit diagram and hardware setup of the Intelligent Lighting Management System using ESP8266.

4.3 Code Snippet

The control program for the Intelligent Lighting Management System using ESP8266 was developed using Arduino IDE. The core logic used for controlling the LED based on the LDR sensor input is shown below. The complete source code is provided in Appendix A.

```
#define LDR D1
```

```
#define LED D2
```

```
int value = digitalRead(LDR);
```

```
Serial.println(value);
```

```
if (value == HIGH) {  
    digitalWrite(LED, HIGH);  
} else {  
    digitalWrite(LED, LOW);  
}
```

CHAPTER 5 – RESULTS AND DISCUSSION

5.1 Output

On power-up, the ESP8266 NodeMCU initializes serial communication and sets the LED to OFF state. During normal daytime conditions, the LDR sensor detects high ambient light intensity and the LED remains OFF, indicating proper functioning of the system.

When the surrounding light intensity decreases (evening or dark conditions), the LDR sensor value changes and the ESP8266 automatically switches the LED ON. This ensures automatic lighting control without any manual intervention.

The Serial Monitor displays real-time LDR sensor readings, which helps in monitoring and verifying the system behavior under different lighting conditions. The system responds quickly and accurately to changes in ambient light.

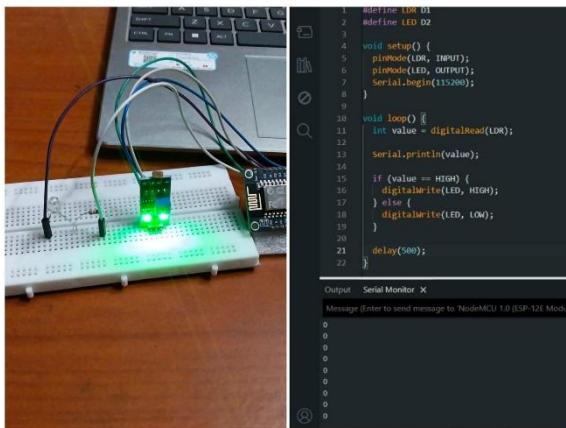


Fig. 5.1: Serial Monitor output showing high light condition (LED OFF)

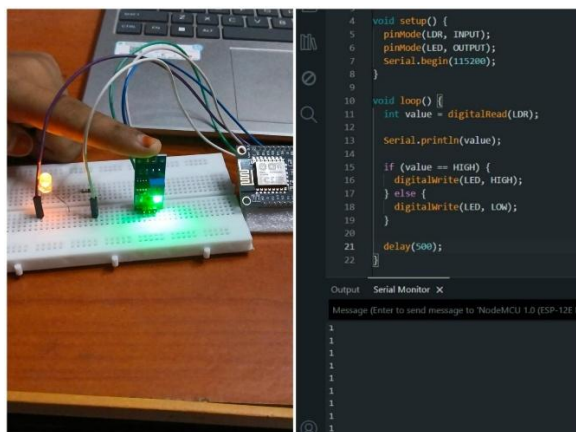


Fig. 5.2: Serial Monitor output showing low light condition (LED ON)

5.2 Testing

Trial	Condition	Observed Sensor Reading	LED Response	Response Time
1	Bright light (daytime)	LOW (0)	OFF	Immediate
2	Partial light (shade)	Changing values	OFF/ON transition	1–2 s
3	Dark condition	HIGH (1)	ON	Immediate
4	Light restored	LOW (0)	OFF	Immediate

Table 5.1: Test readings for different lighting conditions

5.3 Observations

- The system successfully detects changes in ambient light using the LDR sensor.
- The LED automatically turns ON during dark conditions and OFF during sufficient light conditions.
- The ESP8266 processes sensor input quickly and provides stable performance.
- The system operates continuously without manual control, demonstrating effective automation.
- Slight variations in response time may occur due to environmental light changes and sensor positioning.

CHAPTER 6 – CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project successfully demonstrates an Intelligent Lighting Management System using ESP8266 NodeMCU and an LDR (Light Dependent Resistor) sensor. The prototype was implemented on a breadboard and continuously monitors ambient light intensity to control the LED automatically.

The system reliably detects changes in surrounding light conditions and switches the LED ON during low light conditions and OFF during sufficient light conditions. The response is fast, stable, and does not require any manual intervention.

The results confirm that a simple and low-cost sensor-based approach using ESP8266 can effectively automate lighting systems. This helps in reducing unnecessary electricity consumption and improves energy efficiency, making it suitable for smart street lighting and home automation applications.

6.2 Future Scope

1. **Wi-Fi Based Control:** The ESP8266 can be extended to allow remote monitoring and control of the lighting system through a web server or mobile application.
2. **IoT Dashboard Integration:** Sensor data and light status can be displayed in real time using platforms such as Blynk or ThingSpeak for better monitoring.
3. **Automatic Brightness Control:** Instead of simple ON/OFF control, PWM-based dimming can be implemented to adjust light intensity based on surroundings.
4. **Energy Optimization:** Integration of energy-saving algorithms can further reduce power consumption in large-scale deployments.
5. **Sensor Enhancement:** Additional sensors such as motion (PIR) sensors can be added to improve efficiency by activating lights only when movement is detected.
6. **Solar Power Integration:** The system can be powered using solar panels to make it more sustainable and suitable for outdoor street lighting systems.
7. **Smart City Expansion:** The system can be scaled for use in smart city infrastructure for automated street light networks.

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APPENDIX

A.1 Source Code

The following Arduino sketch was uploaded to the ESP8266 NodeMCU to implement the intelligent lighting control logic described in Chapter 3 and Chapter 4.

```
#define LDR D1
#define LED D2

void setup() {
  Serial.begin(115200);
  pinMode(LDR, INPUT);
  pinMode(LED, OUTPUT);
  digitalWrite(LED, LOW);
}

void loop() {
  int value = digitalRead(LDR);

  Serial.print("LDR Value: ");
  Serial.println(value);

  if (value == HIGH) {
    digitalWrite(LED, HIGH);
  } else {
    digitalWrite(LED, LOW);
  }

  delay(500);
}
```

A.2 User Manual

1. Assemble the circuit as per the connections in Table 4.1, ensuring all ground connections are common.
2. Connect the ESP8266 NodeMCU to a computer using a Micro-USB cable.
3. Open Arduino IDE, select the correct ESP8266 board and COM port.
4. Upload the source code given in Appendix A.1.
5. Open the Serial Monitor and set the baud rate to 115200 to view real-time LDR readings.
6. Expose the LDR sensor to different lighting conditions (light/dark) and observe LED response.
7. In dark conditions, the LED turns ON automatically.
8. In bright light conditions, the LED turns OFF automatically.

A.3 Additional Images

Photographs of the assembled hardware prototype and Serial Monitor screenshots captured during testing are included in Figures 3.1,3.2,4.1, 5.1, and 5.2.